

9 Disproof

Each of the following statements is either true or false. If a statement is true, prove it. If a statement is false, disprove it. These exercises are cumulative, covering all topics addressed in Chapters 1-9.

2. For every natural number n , the integer $2n^2 - 4n + 31$ is prime.

Disproof. The statement is false. For a counterexample, let $n = 31$. Then $2n^2 - 4n + 31 = 2(31^2) - 4(31) + 31 = 1829 = 31 \cdot 59$, which is not prime. $\square$

4. For every natural number n , the integer $n^2 + 17n + 17$ is prime.

Disproof. The statement is false. For a counterexample, let $n = 1$. Then $n^2 + 17n + 17 = 1^2 + 17(1) + 17 = 35 = 7 \cdot 5$, which is not prime. $\square$

6. If A, B, C and D are sets, then $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

Proof. Observe the following sequence of equalities.

$$\begin{aligned}(A \times B) \cap (C \times D) &= \{(x, y) : (x, y) \in (A \times B) \wedge (x, y) \in (C \times D)\} \\ &= \{(x, y) : (x \in A \wedge y \in B) \wedge (x \in C \wedge y \in D)\} \\ &= \{(x, y) : (x \in A \wedge x \in C) \wedge (y \in B \wedge y \in D)\} \\ &= \{(x, y) : (x \in A \cap C) \wedge (y \in B \cap D)\} \\ &= \{(x, y) : (x, y) \in (A \cap C) \times (B \cap D)\} \\ &= (A \cap C) \times (B \cap D)\end{aligned}$$

$\square$